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Packaging material

Abstract

A sheet including a pre-storing portion to be a storing portion and a wiring pattern disposed on the sheet are included. The wiring pattern is disposed such that ends of the wiring pattern are disposed on one side along one direction and the wiring pattern is disposed along peripheral edges of the pre-storing portion so as to be along peripheral edges of the storing portion on three sides other than the one side when the storing portion is formed. The sheet is transparent or semitransparent and includes a pre-sealing portion around the pre-storing portion. The wiring pattern is disposed at peripheral edges of the pre-storing portion on the three sides and disposed on the pre-sealing portion.

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Background/Summary

(1) This application is a national phase of International Application No. PCT/JP2022/017244 filed Apr. 7, 2022, which claims priority to Japanese Application No. 2021-070433 filed Apr. 19, 2021, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present technology relates to a packaging material.

BACKGROUND ART

(3) In pharmacotherapy, it is important that patients surely take prescribed medication. The opening detection technology of detecting opening of medicine packaging that helps patients to take medicine as prescribed has been known. For example, Patent Document 1 discloses an opening detection packaging material that includes circuit lines on an enclosing sheet of a press through pack (PTP) packaging material. The PTP packaging material stores tablets separately one by one. The circuit line extends to cross each medicine storing portion and the circuit line is broken together with the enclosing sheet by pressing the storing portion to take out the medicine. The opening of the packaging material can be detected based on a change in an electrical signal caused by the breakage of the circuit pattern.

(4) The PTP packaging material includes a storing section including storing portions of a recessed shape and the enclosing sheet that encloses the storing portions. The storing section generally is generally configured by forming the storing portions having a shape along the shape of the tablets on a thin synthetic resin sheet. Therefore, a target breaking position for opening can be easily specified on the enclosing sheet and the opening can be detected with high accuracy. An aluminum foil that is opaque and has high quality in moisture proof properties, gas barrier properties, and light blocking properties is generally used as the enclosing sheet. Information regarding the stored tablets is printed on the enclosing sheet.

PRIOR ART DOCUMENT

Patent Document

(5) Patent Document 1: Japanese Unexamined Patent Application Publication No. 2020-189636

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(6) As the medicine packaging material, a subdivided bag (one kind of an inner bag) including storing portions each of which can store a dose of powder medicine has been known. The subdivided bag is formed by folding a packaging paper and bonding a peripheral portion. When several kinds of medicines are prescribed, it has been highly demanded to store powder medicine and tablets, which are to be taken at a same time, in each storing portion of the subdivided bag to reduce opportunities to forget to take medicine and take medicine erroneously and to reduce the loads of opening stored bags. However, the subdivided bag is designed with a relatively simple design to be able to store various kinds of medicines, medicines having various shapes, and various number of medicines and therefore, each storing portion has a large area. With such a configuration, it is difficult to specify the opening position and detect the opening.

(7) The present invention was made in view of the above circumstances. An object is to provide a packaging material that is effectively used as a packaging paper for forming a subdivided bag whose opening can be detected.

Means for Solving the Problem

(8) The inventors applied the technology disclosed in Patent Document 1 to a subdivided bag. As illustrated in FIG. 11, a user does not necessarily tear the subdivided bag along a breaking line (may tear the bag along a line X1 or a line X2) when taking a medicine. Therefore, it was found that the opening of the subdivided bag is less likely to be detected appropriately with the technology disclosed in Patent Document 1.

(9) (1) As a result of earnest consideration, a packaging material that is used for forming a subdivided bag including storing portions storing objects is provided with the present technology.

The packaging material includes a sheet including a pre-storing portion to be the storing portion, and a wiring pattern disposed on the sheet. The wiring pattern is disposed such that ends of the wiring pattern are disposed on one side along one direction and the wiring pattern is disposed with respect to the pre-storing portion so as to be along peripheral edges of the storing portion on three sides other than the one side with the storing portion being formed.

(10) (2) In one embodiment of the present technology, in addition to the configuration of (1), the sheet may be transparent or semitransparent and include a pre-sealing portion around the pre-storing portion, and the wiring pattern may be disposed at peripheral edges of the pre-storing portion on the three sides and disposed on the pre-sealing portion.

(11) (3) In one embodiment of the present technology, in addition to the configuration of (1) or (2), the subdivided bag may have a belt shape and the storing portion may include storing portions that are arranged along the one direction. The sheet may have a belt shape and the pre-storing portion includes pre-storing portions that are arranged along the one direction. The wiring pattern may be disposed such that the ends of the wiring pattern are disposed on the one side along the one direction and the wiring pattern is disposed along peripheral edges of the storing portions on the three sides when the storing portions are formed. When the storing portions are formed, the wiring pattern may include a first line that is disposed on one side with respect to the storing portions in a width direction that crosses the one direction and a second line that is disposed on another side in the width direction. The first line and the second line may extend along the one direction. At least one of the first line and the second line may be a common line that is commonly disposed along peripheral edges of the storing portions.

(12) (4) In one embodiment of the present technology, in addition to the configuration of one of (1) to (3), a breaking line may be between the storing portions that are adjacent to each other when the storing portions are formed and the breaking line may include cut portions at intervals. A narrowest portion of the wiring pattern may have a width that is greater than a length of the cut portion.

(13) (5) In one embodiment of the present technology, in addition to the configuration of one of (1) to (4), the packaging material may further include pre-sealing portions around the pre-storing portions. The sheet may include a folding line extending along the one direction in a middle of the sheet with respect to a width direction that crosses the one direction. The sheet may be divided by the folding line into a first section that is on one side and a second section that is on another side with respect to the width direction. With the sheet being folded in half along the folding line and the pre-sealing portion in the first section and the pre-sealing portion in the second section being bonded, the storing portions may be formed.

(14) (6) In one embodiment of the present technology, in addition to the configuration of one of (1) to (5), the packaging material may further include pre-sealing portions around the pre-storing portions. With the sheet and another sheet being overlapped, the pre-sealing portions of the sheets may be bonded to each other and the storing portions may be formed.

Effects of the Invention

(15) According to the present technology, a packaging material that is effectively used as a packaging paper for producing a subdivided bag with whose opening of the bag can be detected is provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a subdivided bag that is formed with a packaging material according to one embodiment and whose opening is being detected.

(2) FIG. 2 is a plan view of the packaging material of FIG. 1.

(3) FIG. 3 is a plan view of a sheet of the packaging material of FIG. 1.

- (4) FIG. 4 is a magnified view of a portion of the subdivided bag near a breaking line.
- (5) FIG. 5 is a block diagram of an opening detector and the subdivided bag in FIG. 1.
- (6) FIG. 6 is a typical view illustrating an example in which opening of the subdivided bag is detected.
- (7) FIG. 7 is a plan view of a packaging material according to another embodiment.
- (8) FIG. 8 is a plan view of a packaging material according to another embodiment.
- (9) FIG. 9 is a plan view of a packaging material according to another embodiment.
- (10) FIG. 10 is a plan view of a packaging material according to another embodiment.
- (11) FIG. 11 is a subdivided bag that is formed with a packaging material according to a reference example and whose opening is being detected.

MODES FOR CARRYING OUT THE INVENTION

First Embodiment

- (12) First, a packaging material according to one embodiment will be generally described with reference to FIGS. 1 to 6. In FIG. 1, opening of a subdivided bag 40 that is formed with using a packaging material 10 is detected. The subdivided bag 40 includes storing portions 42a, 42b, . . . 42n and multiple medicines Med that are to be taken at a same timing are stored in each of the storing portions 42. Breaking lines C1 are between the storing portions 42 and the subdivided bag 40 can be separated into the storing portions 42 along the breaking lines C1. FIG. 2 illustrates the packaging material 10 of this embodiment and is a net view of the subdivided bag 40 in FIG. 1. The packaging material 10 includes a sheet 20 and a wiring pattern 30. Symbols D1 and D2 in the drawings represent a first direction and a second direction, respectively, that are along an elongated direction (one example of one direction) of the packaging material 10. The second direction is opposite from the first direction. Symbols D3 and D4 represent a third direction and a fourth direction, respectively, that are along a width direction. The width direction is perpendicular to the elongated direction. The fourth direction is opposite from the third direction.
- (13) The sheet 20 is a material for producing the subdivided bag 40. The sheet 20 has a rectangular belt-like shape. The sheet 20 of this embodiment is folded in half along a center line with respect to the width direction to produce the subdivided bag 40. A folding line L1 representing a folding position (folding line) at which the sheet 20 is folded is defined at a middle of the sheet 20 with respect to the width direction. In the following, the sheet 20 includes a first section A1 and a second section A2 that are defined by the folding line L1. The first section A1 is on the third direction side and the second section A2 is on the fourth direction side with respect to the width direction.
- (14) Pre-storing portions 21a, 21b . . . 21n are defined along the elongated direction in the first section A1. Pre-storing portions 22a, 22b, . . . 22n are defined along the elongated direction in the second section A2. The number of pre-storing portions 21a, 21b, . . . 21n and pre-storing portions 22a, 22b, . . . 22n is n. n is a natural number that is 1 or greater and n is 6 in the example in FIG. 1. The pre-storing portions 21a, 21b, . . . 21n and the pre-storing portions 22a, 22b, . . . 22n that are adjacent to each other with respect to the width direction, respectively, are disposed symmetrically with respect to the folding line L1 and have a same shape as the storing portions 42a, 42b, . . . 42n. The pre-storing portions 21a, 21b, . . . 21n and the pre-storing portions 22a, 22b, . . . 22n that are arranged in the elongated direction have a same shape as the storing portions 42a, 42b, . . . 42n. Each of the pre-storing portions 21a, 21b, . . . 21n, the pre-storing portions 22a, 22b, . . . 22n, and the storing portions 42a, 42b . . . 42n normally has a same shape. However, the pre-storing portions 21a, 21b, . . . 21n (and the storing portions 42a, 42b, . . . 42n) that are arranged in the elongated direction may have different shapes. As illustrated in FIGS. 2 and 3, the pre-storing portions 21a, 21b, . . . 21n according to this embodiment have a rectangular shape having four round corners. Each of the pre-storing portions 21a, 21b, . . . 21n is surrounded by a pre-sealing portion 24. One end portion of the sheet 20 with respect to the first direction is not the pre-sealing portion 24 but defined as a mount section 28 that is for mounting the packaging material 10 on an opening

detector **50**, which will be described later.

(15) In this description, when each of the pre-storing portions **21a**, **21b**, . . . **21n**, the pre-storing portions **22a**, **22b**, . . . **22n**, and the storing portions **42a**, **42b**, . . . **42n** is not necessarily distinguished from each other, the alphabets at the ends of the symbols are omitted and the components are referred to as the pre-storing portion **21**, the pre-storing portion **22**, and the storing portion **42**. Second lines **32a**, **32b**, . . . **32n** (second lines **32**), third lines **32a**, **32b**, . . . **32n** (second lines **32**), and second terminals **T2a**, **T2b**, . . . **T2n** (second terminals **T2**) may be described similarly.

(16) The pre-sealing portions **24** that are between the pre-storing portions **21** and between the pre-storing portions **22** that are adjacent to each other in the elongated direction have a relatively small dimension measured in the elongated direction and are narrow. The pre-sealing portion **24** that is at a middle of the sheet **20** with respect to the width direction has a relatively small dimension measured in the width direction and is narrow. The pre-sealing portions **24** that are at two edges of the sheet **20** with respect to the width direction have a relatively large dimension measured in the width direction and are wide. Pre-breaking lines **L2** where the breaking lines **C1** are formed are at borders between the pre-storing portions **21** and between the pre-storing portions **22** that are adjacent to each other in the elongated direction (in other words, on the pre-sealing portions **24** that are between the pre-storing portions **21** and between the pre-storing portions **22** that are adjacent to each other in the elongated direction). The folding line **L1** is on a middle of the pre-sealing portion **24**, which is on a middle of the sheet **20** with respect to the width direction, with respect to the width direction. With the sheet **20** being folded along the folding line **L1**, the pre-storing portions **21a**, **21b**, . . . **21n** and the pre-storing portions **22a**, **22b**, . . . **22n** overlap, respectively. With the folded sheet **20** being bonded at the pre-sealing portions **24**, four side portions of each of the pre-storing portions **21a**, **21b**, . . . **21n** and the pre-storing portions **22a**, **22b**, . . . **22n** are closed. Accordingly, the sheet **20** includes closed storing portions **42a**, **42b**, . . . **42n**. The number of storing portions **42** is *n*. The shape of the storing portions **42a**, **42b**, . . . **42n** is not necessarily determined by the shape of the medicine **Med**, which is an object to be stored, and is designed so as to be large enough for the size (a plan shape) of the medicine **Med**, which is an object to be stored.

(17) Various kinds of sheet members made of soft material may be used as the sheet **20** and various sheets that can be thermally bonded with heat pressing are preferably used. One preferable example of the sheet **20** is a layered film that includes semitransparent or transparent thin member (may be a paper) such as glassine and cellophane as a base and an adhesive layer made of thermoplastic resin on a back surface of the base. The base and the adhesive layer are configured as one integral component. Thermoplastic resin may have a softening point or a melting point lower than the heating temperature (for example, about 100° C. to 150° C.) in a sealing process of producing the packaging material such that the thermoplastic resin is softened or melted at the heating temperature. More specifically, examples of the thermoplastic resin may include polyethylene, polypropylene, polystyrene, polyvinyl chloride, and polyethylene terephthalate. The thermoplastic resin is not limited to the above examples. One of the examples may be used as the thermoplastic resin or two or more examples may be combined. With the sheet **20** being semitransparent or transparent as a whole, the objects stored in the subdivided bag **40** can be seen from outside. The sheet **20** has a thickness of about 100 μm or less (60 μm or less, and 40 μm or less typically) and preferably has strength so as to be torn by a human's hand. The sheet **20** may be preferably a resin coated paper having a belt shape, which is a so-called medical packaging paper.

(18) In the present technology, “not having air-permeable properties” means that air-permeability is not observed with the air-permeability test defined by JIS P8117:2009.

(19) The wiring pattern **30** is a component for detecting opening of the subdivided bag **40** that is formed from the packaging material **10**. The wiring pattern **30** is a component that is broken together with the sheet **20** when the sheet **20** is torn to open the storing portion **42**. The wiring pattern **30** has electrically conductive properties and loses the electrically conductive properties (is

disconnected) when the wiring pattern 30 is broken. The wiring pattern 30 of this embodiment is disposed on a surface of the sheet 20. The wiring pattern 30 includes a first line 31, the second lines 32a, 32b, . . . 32n (the second lines 32), and third lines 33a, 33b, . . . 33n (third lines 33). The number of second lines 32 and the number of third lines 33 is n.

(20) The first line 31 extends along the elongated direction in one edge portion of the second section with respect to the width direction. The first line 31 is on the pre-sealing portion 24 and is slightly away from the folding line L1 of the sheet 20 in the fourth direction. The first line 31 extends substantially straight to a second direction-side edge of the n-th pre-storing portion 22n from a first direction-side end. A first terminal T1 is disposed at the first direction-side end (in the mount section 28) of the first line 31. The first terminal T1 is one example of an end of the first line 31. The first line 31 is disposed as close as possible to the folding line L1 (for example, within an area of 5 mm or smaller from the folding line L1). With the first line 31 being disposed on the pre-sealing portion 24, the first line 31 does not block a person from seeing the object stored in the storing portion 42. Therefore, high visibility of the object in the storing portion 42 can be maintained.

(21) The second lines 32 extend along the elongated direction in another edge portion of the second section with respect to the width direction. The second lines 32 are in the pre-sealing portion 24 that is included in a fourth direction-side edge portion of the sheet 20. The second lines 32a, 32b, . . . 32n respectively extend substantially straight to second direction-side edges of the corresponding n-th pre-storing portions 22n from the first direction-side end. The second terminals T2a, T2b, . . . T2n (second terminal T2) are disposed at the first direction-side ends (in the mount section 28) of the second lines 32a, 32b, . . . 32n, respectively. The second terminals T2 are one example of ends of the second lines 32. The second lines 32 of this embodiment are disposed in the pre-sealing portion 24; however, the second lines 32 may be partially disposed on the pre-storing portions 22. With the second lines 32 being disposed on the pre-sealing portion 24, the second lines 32 do not block a person from seeing the object stored in the storing portion 42. Therefore, high visibility of the object in the storing portion 42 can be maintained.

(22) The second lines 32a, 32b, . . . 32n are spaced from each other so as not to be contacted and overlapped with each other. For example, the second line 32a of the second lines 32a, 32b, . . . 32n extends around the storing portion 42a that is at the first direction-side end. The second line 32a is closest to a middle (in other words, closest to the folding line L1) with respect to the width direction and has a relatively small length from the end. The lengths of the second lines 32 from the end become relatively longer and the second lines 32 are disposed on an outer side (in other words, on a fourth direction side) as they are closer to the second line 32n that surrounds the storing portion 42n at a second direction-side end.

(23) The third lines 33a, 33b, . . . 33n connect the first line 31 and the respective second lines 32a, 32b, . . . 32n. The third lines 33 are disposed on the pre-sealing portion 24 that is between the adjacent pre-storing portions 22. Each third line 33 extends to cross the second section A2 along the width direction in a second-direction side edge portion of each n-th pre-storing portion 22n from the first-direction side end. The second lines 32a, 32b, . . . 32n and the third lines 33a, 33b, . . . 33n are, respectively, connected to form an L shape. The first line 31, the second lines 32a, 32b, . . . 32n, and the third lines 33a, 33b, . . . 33n are disposed to surround the storing portions 42a, 42b, . . . 42n, respectively, from three sides (the second direction, the third direction, the fourth direction) except for the first direction. The first line 31 is a common line that is commonly connected to the second lines 32 and the third lines 33 in the wiring pattern 30 such that the storing portions 42 are surrounded by the first line 31, the second lines 32, and the third lines 33.

(24) The widths of the first line 31, the second lines 32, and the third lines 33 are not particularly limited and may be set any values as long as stable electrical conductivity can be confirmed between the first terminal T1 and the second terminals T2. With respect to the improvement in visibility of the stored object, the first line 31, the second lines 32, and the third lines 33 may be

preferably narrow. For increasing the area occupied by the pre-storing portions **21**, **22** (the storing portions **42**) and decreasing the area occupied by the pre-sealing portions **24** in an entire area of the sheet **20**, the lines are preferable to be narrow and the second lines **32** that are arranged parallel to each other in the pre-sealing portion **24** are particularly preferable to be narrower. For example, the width of the first line **31** is preferably 5 mm or less, typically 3 mm or less, and for example, preferably 1.5 mm or less. The width of the second lines **32** is preferably 2 mm or less, typically 1.5 mm or less, and for example, preferably 1 mm or less. The width of the third lines **33** is preferably 5 mm or less, typically 3 mm or less, and for example, preferably 1.5 mm or less. The widths of the first line **31**, the second lines **32**, and the third lines **33** may be preferably 0.5 mm or greater (for example, 0.625 mm or greater).

(25) The breaking lines **C1** are formed at the positions of the pre-breaking lines **L2** when the subdivided bag **40** is formed from the packaging material **10**. As previously described, the pre-breaking lines **L2** extend to cross the sheet **20** in the width direction. Therefore, in the subdivided bag **40**, the breaking lines **C1** may cross the first line **31** and the second lines **32**. For example, as illustrated in FIG. **1**, the breaking lines **C1** extend to cross the first line **31** and the second lines **32b**, . . . **32n**. As illustrated in FIG. **4**, for example, the breaking line **C1** includes cut portions **C2** (one example of a cut portion) each of which has a predefined length **L3** and continuous portions **C3** each having a predefined length **L4**. The cut portions **C2** and the continuous portions **C3** are linearly disposed. It is difficult to form the breaking line **C1** such that the first line **31** or a first portion **32X** is not cut by forming the cut portions **C2**. The width **W** of the narrowest portions of the first line **31** and the second lines **32** are preferably greater than the length **L3** of the cut portion **C2** ($W > L3$). The width **W** and the length **L3** are preferably defined to satisfy the relation of $W \geq 1.25 \times L3$ and more preferably satisfy the relation of $W \geq 1.5 \times L3$. With the width **W** being defined as described above, the cut portions **C2** of the breaking line **C1** can be any position on the pre-breaking line such that the narrowest portions of the first line **31** and the second lines **32** are not completely cut. For example, the width of the narrowest portion of the first line **31** and the width of the narrowest portion of each second line **32** are typically and preferably 0.5 mm or greater, preferably 0.75 mm or greater, and for example, preferably 1 mm or greater. The length **L3** of the cut portions **C2** is preferably 0.5 mm or greater and may be preferably 2 mm or smaller within the range that the above-described relation can be satisfied.

(26) Such a wiring pattern **30** can be formed by supplying an electrically conductive composition on the surface of the sheet **20** in a pattern form and curing the composition. So-called electrically conductive ink that is obtained by dispersing electrically conductive particles in a vehicle including a binder can be used as the electrically conductive composition. Various kinds of particles (may be formed into powders) of material having good electrically conductive properties such as gold, silver, copper, platinum, aluminum, an alloy thereof, and carbon black may be used as the electrically conductive particles. The electrically conductive particles may have various kinds of shapes such as a spherical shape, a flat plate shape, a polyhedron shape, and a rod shape. The vehicle typically includes a binder and a dispersion medium and the composition and physical properties are adjusted according to the supplying method to be performed.

(27) The method of supplying the electrically conductive component is not particularly limited and may be a supplying method with a dispenser and various kinds of printing methods such as letterpress printing, calendar printing, offset printing, ink jetting printing, and screen printing. A dry-curing method, a heat-curing method, and a sintering method may be used as the method of curing the electrically conductive component as appropriate according to the physical properties of the electrically conductive component to be used.

(28) At least a portion of the wiring pattern **30** is disposed on the pre-sealing portion **24** and is exposed to a predefined heating temperature (determined according to the thickness of the sheet **20** and the material of the adhesive layer, and about 100° C. to 150° C., for example) at which the packaging material **10** is subjected to the heating and pressing process to form the bag. The wiring

pattern **30** may preferably have heat resistant properties to keep the physical properties such as the shape and electrically conductive properties during the heating and pressing process for forming the bag. Therefore, the wiring pattern **30** may be preferably made of the electrically conductive composition having heat resistant properties. Examples of such an electrically conductive composition having heat resistant properties include screen inks made by TOYO INK such as RAFS059, RAFS007, RAFS076, RAFS090, and RAFS110S. However, the electrically conductive composition is not necessarily limited to the above examples.

(29) The packaging material **10** according to this embodiment includes opening order marks **25** and a marking **26** on the surface of the sheet **20** in addition to the wiring pattern **30**. The opening order marks **25** and the marking **26** may be formed with the electrically conductive component that is same as or similar to that of the wiring pattern **30** at the same time as the wiring pattern **30** is formed. The material of the marks may not be limited to the above one.

(30) The opening order mark **25** is provided on each of the pre-storing portions **22a**, **22b**, . . . **22n** and represents the order by which the storing portions **42** are to be opened. The opening order marks **25** may be symbols representing numbers or the order. Preferable examples of the opening order marks **25** may include Arabic numerals, Roman numerals, numerals with Chinese characters, Hiragana characters, alphabets, and Chinese characters. Arabic numerals are provided on the packaging material **10** of this embodiment as the opening order marks **25**. The opening order marks **25** are numbered in an ascending order from the pre-storing portion **22n** that is on the second direction-side end to the pre-storing portion **22a** that is on the first direction-side end. This guides a user to open the storing portions **42** of the subdivided bag **40** in a correct order.

(31) The marking **26** is a mark that represents a reference position that is used when the packaging material **10** or the subdivided bag **40** is to be processed with other devices. A type and a shape of the marking **26** are not particularly limited as long as a sheet-shaped object can be positioned at a predefined position. Various marks that can be read by a device to be used such as a packaging device or the opening detector **50** can be used as the marking **26**. The marking **26** may be a cross-shaped mark that can be read by an infrared laser, for example. A portion of the wiring pattern **30** or an entire wiring pattern **30** may be used as the marking **26**. With the packaging material **10** including such a marking **26**, the sheet **20** can be bonded with less deviation from the pre-sealing portions **24** when the bag is formed from the packaging material **10**. By forming the marking **26** on the sheet **20** at the same time as the wiring pattern **30** is formed, the relative position of the marking **26** and the wiring pattern **30** can be adjusted strictly and the sheet **20** is effectively bonded such that the wiring pattern **30** is placed on the pre-sealing portion **24** with reference to the marking **26**. With the packaging material **10** including such a marking **26**, a connection terminal section **51**, which will be described later, the first terminal **T1** and the second terminals **T2** can be contacted more surely when the subdivided bag **40**, **10** is connected to the opening detector **50**.

(32) The packaging material **10** having such a configuration is folded in half along the folding line **L1** such that the surface faces outside and the folded packaging material **10** is inserted in a known packaging device (not illustrated). In the packaging device, the bag is formed with the packaging material **10** being transferred in a forwarding direction that is along the elongated direction. Specifically, in the packaging device, the folded packaging material **10** is opened and the opposing sheet **20** portions are closed and the pre-sealing portions **24** of the sheet **20** are heated and pressed to form the storing portions **42**. In the packaging device, when the sheet **20** portions are closed, predefined medicine **Med** is put in each space between the opposing sheet **20** portions. Thus, multiple kinds of medicines **Med** to be taken one time are stored in each of the storing portions **42** and the storing portions **42** are sealed. Then, in the packaging device, the breaking lines **C1** are formed in the bonded portions, which are between the storing portions **42**, by cutting portions of the bonded portions along the width direction with a perforation blade. Thus, the subdivided bag **40** in which a plurality of medicines **Med** are stored in each storing portion is prepared.

(33) The opening detector **50** is a device for detecting opening of the subdivided bag **40** that is

formed with using the packaging material **10**. As illustrated in FIG. 5, for example, the opening detector **50** includes the connection terminal section **51**, a control section **52**, an operation section **53**, an informing section **54**, and a battery arrangement section **55**.

(34) The connection terminal section **51** includes connection terminals that are connected to the first terminal **T1** and the second terminals **T2** of the subdivided bag **40**. The connection terminal section **51** is configured to be electrically connected to the first terminal **T1** and the second terminals **T2** of the wiring pattern **30** of the subdivided bag **40** when the first direction-side end portion of the subdivided bag **40** is mounted in a predefined mount portion of the opening detector **50**.

(35) The control section **52** is configured to detect opening of each of the storing portions **42** of the subdivided bag **40**. More specifically, a microcomputer that includes an interface (I/F), a central processing unit (CPU), a read only memory (ROM), a random access memory (RAM), a storing section **M** storing various kinds of information, and a timer **T** having a counter function is configured as the control section **52**. Various kinds of information are transmitted through the interface. The CPU performs operations according to the commands from the control program. The ROM stores programs that are performed by the CPU. The RAM is used as a working area for developing the program. The ROM, for example, stores a computer program, a data base, and a table that are used for various kinds of processing performed by the opening detector **50**. The storing section **M**, for example, stores ID information related to a user or the opening detector **50**, information related to the time when the medicine is to be taken, information used for the various processing performed by the opening detector **50**, and information related to the results of the opening detection.

(36) A user sends instructions related to driving to the opening detector **50** via the operation section **53**. The operation section **53** of this embodiment includes push buttons and with any one of the push buttons being pressed, the instruction that is predefined corresponding to the pressed push button is sent to the control section **52**. For example, by operating one of the push buttons of the operation section **53**, the instruction of performing the opening detection of the subdivided bag **40** is sent to the control section **52**.

(37) The informing section **54** is a component for informing a user of information related to the driving of the opening detector **50**. The informing section **54** of this embodiment includes a buzzer and is configured to make alarm sound with one or more informing patterns. The informing section **54** is configured to make alarm sound with a predefined informing pattern to inform of the time to take medicine when it is the time predefined by a user to take medicine. If the position where the subdivided bag **40** is mounted in the opening detector **50** by a user is not appropriate, alarm sound that informs of a mounting error with a predefined informing pattern can be made. The informing section **54** is configured to make alarm sound to with a predefined informing pattern to inform of completion of detection when the opening detector **50** finishes the opening detection of the subdivided bag **40**.

(38) The battery arrangement section **55** is a component for receiving a battery, which is a driving power source for driving the opening detector **50**, and is electrically connected to the control section **52** and the informing section **54** to supply power.

(39) Such a subdivided bag **40** formed with using the packaging material **10** can be used in a medication instruction system illustrated in FIG. 6, for example. Specifically, when the subdivided bag **40** is connected to the opening detector **50**, the control section **52** sends a signal to one terminal (for example, the first terminal **T1**) that is connected to the connection terminal section **51** and detects the signal via the connection terminal section **51** that is connected to another terminal (for example, the second terminal **T2**) and thus detects an electrically conductive state between the first terminal **T1** and the second terminal **T2**. In FIG. 6, no storing portions **42** (the number of storing portions **42** is **n**) of the user **P1**'s subdivided bag **40** is opened and the first terminal **T1** and the second terminals **T2** are electrically conductive. Then, the control section **52** of the user **P1**'s

opening detector **50** detects the electrically conductive state between the first terminal **T1** and each of the second terminals **T2** and stores the detection results and the time information in the storing section **M**. With respect to the user **P2**'s subdivided bag **40**, the medicine is taken two times and the two storing portions **42** are cut away. If the n -th storing portion **42 n** from the first direction-side end is opened, at least portions of the first line **31** and the second line **32 n** that surround the storing portion **42 n** and are disposed at a peripheral portion of the storing portion **42 n** are broken together with the sheet **20**. As a result, the electrical connection between the first terminal **T1** and the second terminals **T2** is disconnected. More specifically, the first terminal **T1** is electrically connected to the second terminals **T21** to **T24** but the electrical connection between the first terminal **T1** and each of the second terminals **T25**, **T26** is disconnected. The control section **52** of the user **P2**'s opening detector **50** detects the disconnection between the first terminal **T1** and the second terminals **T2** and stores the detection result and the time information in the storing section **M**.

(40) The information related to the opening detection results obtained by the opening detector **50** can be, for example, transmitted to a management server **60** of the medication instruction system with wired connection or wireless connection. Accordingly, the user **P1**, **P2** can receive appropriate medication instructions based on their medication records.

(41) The packaging material **10** previously described includes the sheet **20** including the pre-storing portions **21**, **22** that are to form the storing portions **42** storing the medicine **Med** (stored object) and the wiring pattern **30** disposed on the sheet **20**. The wiring pattern **30** is disposed such that ends of the wiring pattern **30** are on the first direction side (one side) along the elongated direction (one direction). The wiring pattern **30** is disposed with respect to the pre-storing portion **22** so as to extend along three peripheral edges of the storing portion **42** except for the first direction-side edge when the storing portion **42** is formed from the pre-storing portion **22**.

(42) According to the above configuration, the wiring pattern **30** is disposed to surround each of the storing portions **42** from three directions. Two ends of the line that surrounds each storing portion are disposed at the first direction-side end of the sheet **20** and can be connected to a target easily with the opening detector **50**. Accordingly, the physical change can be detected. When the sheet **20** of the subdivided bag **40** that is formed with using the packaging material **10** is torn to take out the medicine **Med** from any one of the storing portions **42**, the sheet **20** may be broken with various ways as described below. The storing portion **42** may be torn completely along a line crossing the storing portion **42** along the width direction (refer to a line **C2** in FIG. **1**). The storing portion **42** may be torn from one side to an intermediate portion with respect to the width direction. The storing portion **42** may be torn obliquely from one side to the second direction-side end with respect to the width direction (refer to a line **C3** in FIG. **1**). In any of the above breaking ways, the wiring pattern **30** is broken together with the sheet **20**. The physical change (for example, the electrical conductive properties of the wiring pattern **30** that surrounds the storing portion **42**) occurs due to the breakage of the wiring pattern **30**. Based on the physical change, the breakage of the sheet **20** in the storing portion **42** (in other words, the opening of the subdivided bag **40**) can be detected.

(43) In the above packaging material **10**, the sheet **20** is transparent or semitransparent and includes the pre-sealing portions **24** around the pre-storing portions **21**, **22**. The wiring pattern **30** is arranged on the pre-sealing portions **24** that are peripheral edges of the pre-storing portion **22** on the three direction sides (the second direction, the third direction, the fourth direction). According to such a configuration, the pre-storing portions **21**, **22** and the storing portions **42** are transparent or semitransparent and the wiring pattern **30** is arranged so as not to overlap the pre-storing portions **21**, **22** and the storing portions **42**. Accordingly, the subdivided bag **40** that is formed with using the packaging material **10** can have a configuration in which the medicine **Med** stored in the storing portion **42** is less likely to be blocked by the wiring pattern **30** and the medicine **Med** can be seen from outside. This is particularly effective for the subdivided bag **40** with which several kinds of medicines **Med** are packed in each storing portion **42**.

(44) With respect to the packaging material **10**, the subdivided bag **40** has a belt shape and includes the storing portions **42** that are arranged along the elongated direction (the one direction). The sheet **20** has a belt shape and includes the pre-storing portions **21**, **22** that are arranged along the elongated direction. The wiring pattern **30** is disposed such that ends of the wiring pattern **30** are disposed on the first direction side (one direction) along the elongated direction. The wiring pattern **30** extends along three peripheral edges (the second direction side, the third direction side, the fourth direction side) of each of the storing portions **42** with respect to the pre-storing portions **21**, **22**. The wiring pattern **30** includes the first line **31** and the second lines **32**. The first line **31** is disposed on the third direction-side (one side with respect to the width direction that crosses the elongated direction) with respect to the storing portions **42** and extends along the elongated direction. The second lines **32** are disposed on the fourth direction-side (another side with respect to the width direction) with respect to the storing portions **42** and extend along the elongated direction. The first line **31** is a common line that is provided commonly for the storing portions **42** and extends along the peripheral edges of the storing portions **42**. According to such a configuration, unlike the second lines **32a**, **32b**, . . . **32n** that are arranged parallel to each other, the first line **31** needs only a small area to be arranged. Therefore, with the wiring pattern **30** being arranged on the pre-sealing portions **24**, the ratio of the areas of the pre-sealing portions **24** to the sheet **20** can be reduced. As a result, the ratio of the areas of the pre-storing portions **21**, **22** to the sheet **20** (namely, the ratio of the areas of the storing portions **42** to the subdivided bag **40**) can be increased and this improves the storage ability and the visibility of the stored objects.

(45) With respect to the subdivided bag **40** formed with using the above packaging material **10**, the breaking lines **C1** including cut portions at intervals extend between the adjacent storing portions **42** along the width direction. In the packaging material **10**, the width of the narrowest portion of the wiring pattern **30** is greater than the length of the cut portion of the breaking line **C1**. With such a configuration, when the breaking line **C1** is formed in the subdivided bag **40** that is formed with using the packaging material **10**, the wiring pattern **30** is less likely to be broken by forming the breaking line **C1**. When one storing portion **42n** is separated from the subdivided bag **40**, the storing portions **42** that are adjacent to each other with sandwiching the breaking line **C1** are easily separated and other storing portion **42(n-1)** is less likely to be erroneously opened.

(46) The packaging material **10** includes the pre-sealing portions **24** around the pre-storing portions **21**, **22**. The sheet **20** includes the folding line **L1** that extends along the elongated direction in a middle with respect to the width direction. The sheet **20** is divided by the folding line **L1** into the first section **A1** that is on one side and the second section **A2** that is on another side with respect to the width direction. With the sheet **20** being folded in half along the folding line **L1**, the pre-sealing portions **24** in the first section **A1** and the pre-sealing portions **24** in the second section **A2** are bonded to each other and the storing portions **42** are formed. According to such a configuration, the subdivided bag **40** is easily formed from one sheet **20**.

Second Embodiment

(47) A packaging material **110** according to a second embodiment will be described with reference to FIG. 7. The packaging material **110** according to the second embodiment includes pre-storing portions **122**. The pre-storing portion **122** has a shape that is formed by connecting the pre-storing portion **21** and the pre-storing portion **22** of the first embodiment in the width direction. The packaging material **110** includes two circular holes **126** in the mount section **28**. The circular holes **126** are arranged symmetrically with respect to the folding line **L1**. Other configurations are similar to those of the first embodiment and the configurations, operations, and effects similar to those of the first embodiment will not be described.

(48) Specifically, the pre-storing portions **122** of the packaging material **110** have a rectangular shape that is elongated in the width direction. Therefore, the pre-sealing portion extending in the elongated direction is not provided in a middle with respect to the width direction. The folding line **L1** and a first line **131** extend to cross the pre-storing portions **122**. The pre-storing portion **122** is

disposed symmetrically with respect to the folding line L1. The pre-storing portion **122** is about twice as large as the storing portion.

(49) The packaging material **110** includes pre-sealing portions **124** around the pre-storing portions **122**. A sheet **120** includes the folding line L1 that extends along the elongated direction in a middle with respect to the width direction. The sheet **120** is divided by the folding line L1 into the first section A1 that is on one side and the second section A2 that is on another side with respect to the width direction. With the sheet **120** being folded in half along the folding line L1, the pre-sealing portions **124** in the first section A1 and the pre-sealing portions **124** in the second section A2 are bonded to each other and the storing portions **42** are formed. The two circular holes **126** are overlapped with each other and one circular hole is formed. A columnar projection included in the opening detector **50** can be inserted through the circular hole.

(50) According to such a configuration, with the sheet **120** being folded in half, three peripheral edge portions of the pre-storing portion **122** except for one peripheral edge close to the folding line L1 are bonded and the storing portion **42** can be enclosed. This simplifies a bonding process of bonding the sheet **120** and reduces the area of the pre-sealing portions **124**. Eventually, a cost of the packaging material **10** for forming the subdivided bag **40** can be reduced. Furthermore, with the circular holes **126** being included in the mount section **28**, the positioning with respect to the opening detector **50** becomes easy and the first terminal T1 and the second terminals T2 of the packaging material **110** can be surely connected to the connection terminal section **51** of the opening detector **50**.

Third Embodiment

(51) A packaging material **210** according to a third embodiment will be described with reference to FIG. 8. The packaging material **210** of the third embodiment includes a wiring pattern **230** and a wiring arrangement of the wiring pattern **230** differs from that of the first embodiment and the second embodiment. Other configurations are similar to those of the first embodiment and the second embodiment and the configurations, operations, and effects similar to those of the first embodiment and the second embodiment will not be described.

(52) Specifically, the wiring pattern **230** of the packaging material **210** is configured such that positions of a first line **231** and second lines **232** are opposite from the positions of the second embodiment with respect to the width direction. With such a configuration, the frequency that the second lines **232** are disposed on a pre-sealing portion **224** can be reduced. Accordingly, when the subdivided bag **40** is formed from the packaging material **210**, the wiring pattern **230** is less likely to be thermally affected.

Fourth Embodiment

(53) A packaging material **310** according to a fourth embodiment will be described with reference to FIG. 9. The packaging material **310** of the fourth embodiment includes a wiring pattern **330** and a wiring arrangement of the wiring pattern **330** differs from that of the first to third embodiments. Other configurations are similar to those of the first to third embodiments and the configurations, operations, and effects similar to those of the first to third embodiments will not be described.

(54) Specifically, in the first to third embodiments, the first lines **31**, **131**, **231** and the second lines **32**, **132**, **232** are on the same side with respect to the folding line L1 on the sheets **20**, **120**, **220**. In the packaging material **310** according to the fourth embodiment, some second lines **332** and third lines **333** are disposed on one side (third direction-side, in an A1 section) and another second lines **332** and third lines **333** are on another side (fourth direction-side, in an A2 section) with respect to the folding line L1 in the width direction of the sheet **20**. For example, the positions of the second lines **332** and the third lines **333** disposed around the respective pre-storing portions **322** that are adjacent to each other are opposite from each other with respect to the folding line L1. According to such a configuration, with the second lines **332** being disposed on the pre-sealing portion **324**, a width dimension of the pre-sealing portions **324** at two edges of the sheet **320** with respect to the width direction can be reduced. The width of the second lines **332** or a space between the adjacent

second lines **332** can be increased.

Fifth Embodiment

(55) A packaging material **410** according to a fifth embodiment will be described with reference to FIG. **10**. The fifth embodiment includes the packaging material **410** and another packaging material **510** to form the subdivided bag **40**. The size of the packaging material **410** differs from that of the first to fourth embodiments. Other configurations are similar to those of the first to fourth embodiments and the configurations, operations, and effects similar to those of the first to fourth embodiments will not be described.

(56) Specifically, in the first embodiment, for example, the dimension of the sheet **20** measured in the width direction is about twice as the dimension of the pre-storing portion **21** (namely, the dimension of the subdivided bag **40** and the storing portion **42**) measured in the width dimension such that the sheet **20** is folded in half to form the subdivided bag **40**. The packaging material **410** of the fifth embodiment has a shape obtained by cutting the sheet **20** of the first embodiment along the folding line **L1**. In forming the subdivided bag **40**, the packaging material **410** and the sheet **520** (which may be same as the sheet **420**) without including the wiring pattern **430** are bonded at pre-sealing portions **424**, **524** that are included in four edge portions of the respective pre-storing portions **421**, **521** and the storing portions **42** are formed. According to such a configuration, the wiring pattern **430** need not be disposed on the packaging material **510** and therefore, a conventional belt-shaped packaging paper can be used. Accordingly, a cost of the packaging material **10** for forming the subdivided bag **40** can be reduced.

(57) Preferred embodiments of the present disclosure are described with reference to the drawings; however, the technology described herein is not limited to the above embodiments. It is obvious for those having skills in the field of the technology described herein to achieve various modifications or alternations within the technological scope of the present invention and such modifications and alternations are included in the technological scope of the present disclosure. For example, the following embodiments may be included in the technical scope of the present invention.

(58) (1) In the above embodiments, the sheet includes multiple pre-storing portions and has a belt shape. However, the sheet may include only one pre-storing portion and may not have a belt shape. In such a configuration, the sheet may have a wide shape that is greater in the width direction than is perpendicular to the one direction than in the one direction.

(59) (2) In the above embodiments, a thin member (which may be a paper) including fibers such as glassine and cellophane is used as a base; however, the base may not necessarily include fibers but may be a synthetic resin sheet. The synthetic resin sheet used as the base may preferably have heat resistant properties with respect to the heating temperature in the forming of the bag and the sealing.

(60) (3) In the above embodiments, the wiring pattern **30** is disposed on the surface of the sheet **20** (the surface of the base). The wiring pattern may be disposed on a back surface of the sheet **20** (a surface of thermoplastic resin disposed on the base) as long as the opening can be detected with the wiring pattern that is disposed at a peripheral edge of one storing portion when the storing portion is opened.

(61) (4) In the first and fifth embodiment, the wiring patterns **30**, **430** are entirely disposed on the pre-sealing portions **24**, **424**. However, when the bag is actually formed, a portion of the wiring pattern **30**, **430** is allowed to be disposed outside the sealing portions. A portion of the wiring pattern **30**, **430** may be disposed outside the pre-sealing portion **24**, **424** as long as the object of the present technology is achieved.

EXPLANATION OF SYMBOLS

(62) **10**, **110**, **210**, **310**, **410**: packaging material, **20**, **120**, **220**, **320**, **420**: sheet, **21**, **21a**, **21b** . . . **21n**, **22a**, **22b** . . . **22n**, **122**, **222**: pre-storing portion, **24**, **124**, **224**, **324**, **424**: pre-sealing portion, **30**, **130**, **230**, **330**, **430**: wiring pattern, **31**, **131**, **231**, **331**, **431**: first line, **32**, **132**, **232**, **332**, **432**: second line, **33**, **133**, **233**, **333**, **433**: third line, **40**: subdivided bag, **42**, **42a**, **42b** . . . **42n**: storing

portion, 50: opening detector, C1: breaking line, C2: cut portion (cut line), L1: folding line, L2: pre-breaking line, L3: length (cut length), L4: length, Med: medicine (stored object)

Claims

1. A packaging material used for forming a subdivided bag including a storing portion that stores an object, the packaging material comprising: a sheet including a pre-storing portion to be the storing portion; and a wiring pattern disposed on the sheet, wherein the wiring pattern is disposed such that ends of the wiring pattern are disposed on one side along one direction and the wiring pattern is disposed with respect to the pre-storing portion so as to be along peripheral edges of the storing portion on three sides other than the one side with the storing portion being formed, the subdivided bag has a belt shape and the storing portion includes storing portions that are arranged along the one direction, the sheet has a belt shape and the pre-storing portion includes pre-storing portions that are arranged along the one direction, the wiring pattern is disposed such that the ends of the wiring pattern are disposed on the one side along the one direction and the wiring pattern is disposed along peripheral edges of the storing portions on the three sides when the storing portions are formed, the wiring pattern includes a first line that is disposed on one side with respect to the storing portions in a width direction that crosses the one direction and a second line that is disposed on another side in the width direction when the storing portions are formed, the first line and the second line extend along the one direction, and at least one of the first line and the second line is a common line that is commonly disposed along the peripheral edges of the storing portions.
2. The packaging material according to claim 1, wherein the sheet is transparent or semitransparent and includes a pre-sealing portion around the pre-storing portion, and the wiring pattern is disposed at peripheral edges of the pre-storing portion on the three sides and disposed on the pre-sealing portion.
3. The packaging material according to claim 1, wherein a breaking line is between the storing portions that are adjacent to each other when the storing portions are formed, the breaking line includes cut portions at intervals, and a narrowest portion of the wiring pattern has a width that is greater than a length of one of the cut portions.
4. The packaging material according to claim 1, further comprising pre-sealing portions around the pre-storing portions, wherein the sheet includes a folding line extending along the one direction in a middle of the sheet with respect to a width direction that crosses the one direction, the sheet is divided by the folding line into a first section that is on one side and a second section that is on another side with respect to the width direction, and with the sheet to be folded in half along the folding line and the pre-sealing portion in the first section and the pre-sealing portion in the second section to be bonded, the storing portion is formed.
5. The packaging material according to claim 1, further comprising pre-sealing portions around the pre-storing portions, wherein with the sheet and another sheet being overlapped, the pre-sealing portions of the sheets are to be bonded to each other and the storing portion is to be formed.
6. A subdivided bag having at least four edges, the subdivided bag comprising: a storing portion that stores an object; and a wiring pattern disposed such that ends of the wiring pattern are disposed on a first edge of the at least four edges, the wiring pattern extending around the storing portion and along three edges of the at least four edges except for the first edge, wherein the subdivided bag has a belt shape and the storing portion includes storing portions that are continuously arranged in an elongated direction, the wiring pattern is disposed such that the ends of the wiring pattern are disposed on the first edge along the elongated direction and the wiring pattern is disposed along peripheral edges of the storing portions on the three edges, the wiring pattern includes lines that are provided corresponding to the storing portions, respectively, the wiring pattern includes a first one of the lines that is disposed on one side with respect to the storing portions in a width direction that crosses the elongated direction and a second one of the lines that is disposed on another side in the

width direction, the first line and the second line extending along the elongated direction, and at least one of the first one of the lines and the second one of the lines is a common line that is commonly disposed along the peripheral edges of the storing portions.

7. The subdivided bag according to claim 6, wherein each of the storing portions has at least four edges including a first edge that is parallel to the first edge of the subdivided bag, the storing portions are continuously arranged such that the first edges of the storing portions are parallel to each other and the storing portions are able to be separated from each other, two ends of each of the lines are disposed on the first edge of the subdivided bag, and each of the lines extends along the three edges of each of the storing portions except for the first edge of each of the storing portions.

8. The subdivided bag according to claim 7, wherein the at least four edges of each of the storing portions include a second edge that is opposite the first edge and a third edge and a fourth edge that are between the first edge and the second edge and extend parallel to each other, the storing portions at least include a first storing portion that is closest to the first edge of the subdivided bag and a second storing portion that is next to the first storing portion in the elongated direction, the lines include a first surrounding line that surrounds the first storing portion and extends from one of the two ends along the third edge, the second edge, and the fourth edge of the first storing portion to another one of the two ends, and the lines include a second surrounding line that extends from one of the two ends along the third edge of the first storing portion, the third edge, the second edge, and the fourth edge of the second storing portion, and the fourth edge of the first storing portion to another one of the two ends.
